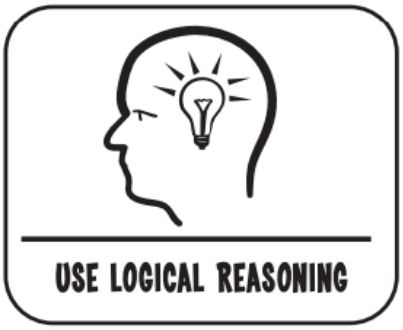
STRATEGY 7:

Use Logical Reasoning

Logical reasoning is a way to help students organize data and use the process of elimination to solve problems. Logic boxes and logic lines (see page 80) are helpful tools for organizing facts and using the process of elimination.



SAMPLE PROBLEM 1

Skippy, Steve, and Stanley are on their way to a family reunion. One is traveling by hot-air balloon, another by blimp, and the third by pogo stick. Who's on the pogo stick? Use these clues to figure it out:

- Steve is afraid of blimps.
- Stanley is afraid of pogo sticks.
- Skippy loves balloons.

Solution

From the clues, we know that Steve won't travel by blimp. Put an X in the logic box next to Steve under Blimp. Stanley is afraid of pogo sticks so put an X next to Stanley under Pogo Stick. Skippy loves balloons so put a check next to Skippy under Balloon, and X out Pogo Stick and Blimp. X out the rest of the Balloon column. If Steve isn't in the balloon or blimp, he must be on the pogo stick. **Answer:** Steve is on the pogo stick.

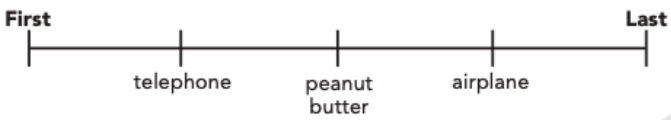
	Blimp	Pogo Stick	Balloon
Steve	✗	✓	✗
Stanley	✓	✗	✗
Skippy	✗	✗	✓

SAMPLE PROBLEM 2

The airplane was invented after the telephone and peanut butter. George Washington Carver invented peanut butter after Alexander Graham Bell gave us the telephone. Which invention came first?

Solution

To solve, draw a logic line. We know that peanut butter goes to the left of the airplane and that the telephone goes to the left of peanut butter. **Answer:** The telephone was the first to be invented of the three.

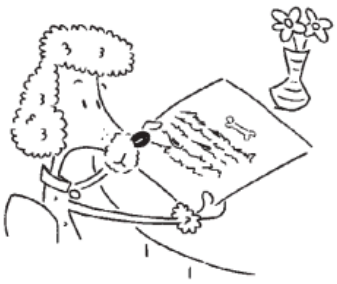


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99 Logic and Reasoning

It's time for the final race of the Snail Olympics. The American snail finishes before the German snail, but after the Irish snail. The Irish snail finishes before the French snail, but after the Swedish snail. Who won the race? Use a logic line to find out.

100 Logic and Reasoning



Murray, Minnie, and Molly Morton are at the Mama Meatball restaurant with their pet poodle, Pepper. They ordered a plate of fried chicken, a seaweed salad, a bucket of bones, and a chicken potpie. Who ordered what? Use the clues and a logic box to find out:

- Murray can't eat fried food.
- Molly is a vegetarian.
- Pepper hates chicken.

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STRATEGY 1:

Choose an Operation



Even the most straightforward problem requires a mathematical operation to solve it. The question is, which one? After examining the information presented in a problem, students must decide which operation (addition, subtraction, multiplication, or division) they should use to solve it. Instruct them to read the problem and look for key words and phrases, such as "all together" or "more than" that may help move them in the right direction when choosing an operation. Then have them write an actual equation and solve.

SAMPLE PROBLEM

Uncle Otto ate a whole pumpkin pie in 21 minutes flat. Cousin Orville ate a whole pie in $\frac{1}{3}$ of that time. How long did it take Orville to eat the pie?

Solution

Otto ate the pie in 21 minutes. Orville's time was $\frac{1}{3}$ of that. One operation we could use to figure out Orville's time is division. With the information in the problem, we can create an equation:

$$21 \div 3 = ?$$

Answer: It took Orville 7 minutes to eat the whole pie.

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24 Fractions and Decimals

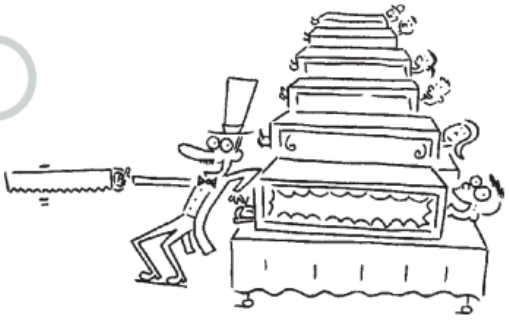


It's feeding time at the Bonks Zoo. The monkeys get 6,000 bananas. The bunnies get half as many carrots as the monkeys get bananas. And the zookeepers get $\frac{1}{3}$ as many peanut-butter-and-jellyfish sandwiches as the bunnies get carrots. How many sandwiches do the zookeepers get?

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FRACTIONS AND DECIMALS

25 Fractions and Decimals



Magico the magician has 6 volunteers from the audience. He plans to saw each one in half. And then he will saw the halves in half. How many pieces of volunteers will he have at the end of the trick?